Experiments continued

What matters for compatibility recovery?

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Recap

Results

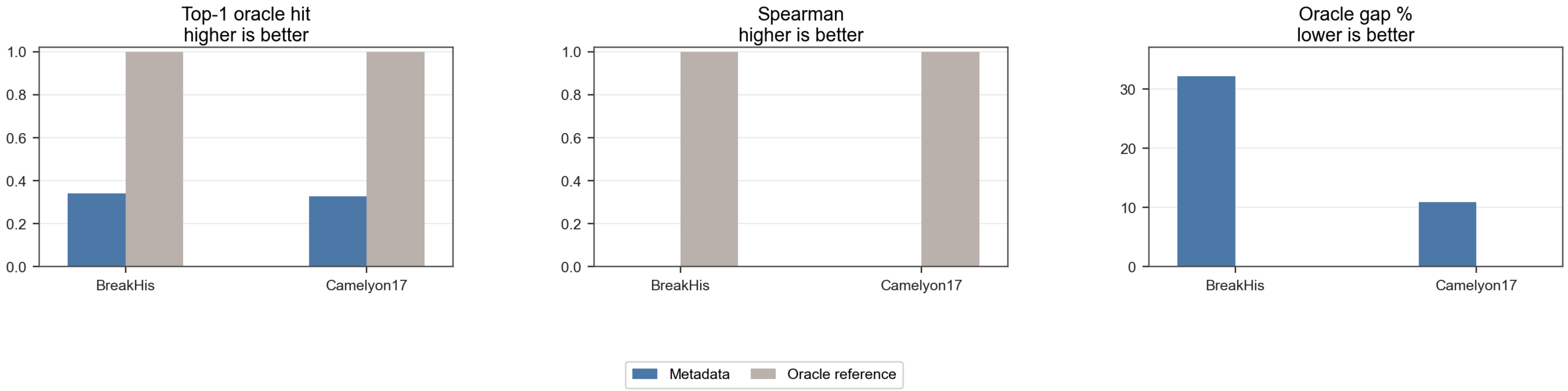
- Source-only learned utility family
- Source-only AE family
- Target-local direct support-NELBO

Key takeaways

Recap

Metadata-Baseline

Metadata similarity does not suffice for compatibility routing



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Where we stopped last time

Key points

- Metadata routing changes expert selection but does not reliably improve utility
- Learned proxy models and metadata-conditioned variants did not recover utility-aligned rankings
- The bottleneck was localized to compatibility estimation, not the routing mechanism itself

The new question

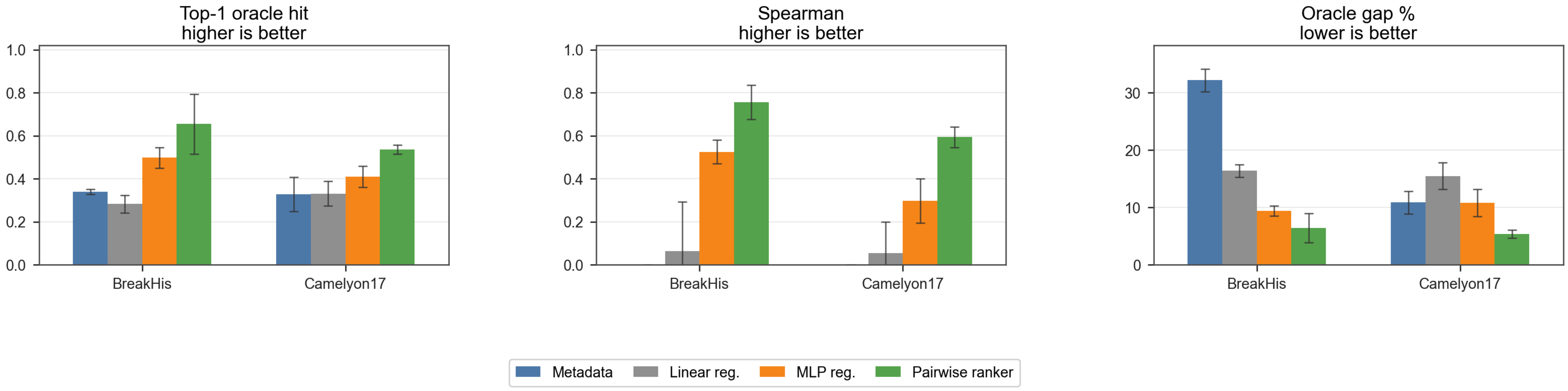
How can we estimate compatibility without leakage?

- Learned utility predictors
  - Pairwise ranker
- Source-only AE confidence family
  - Raw AE confidence
  - AE utility calibrator
- Target-local direct support-NELBO

Results

Result A

Learned utility is not adoption-eligible



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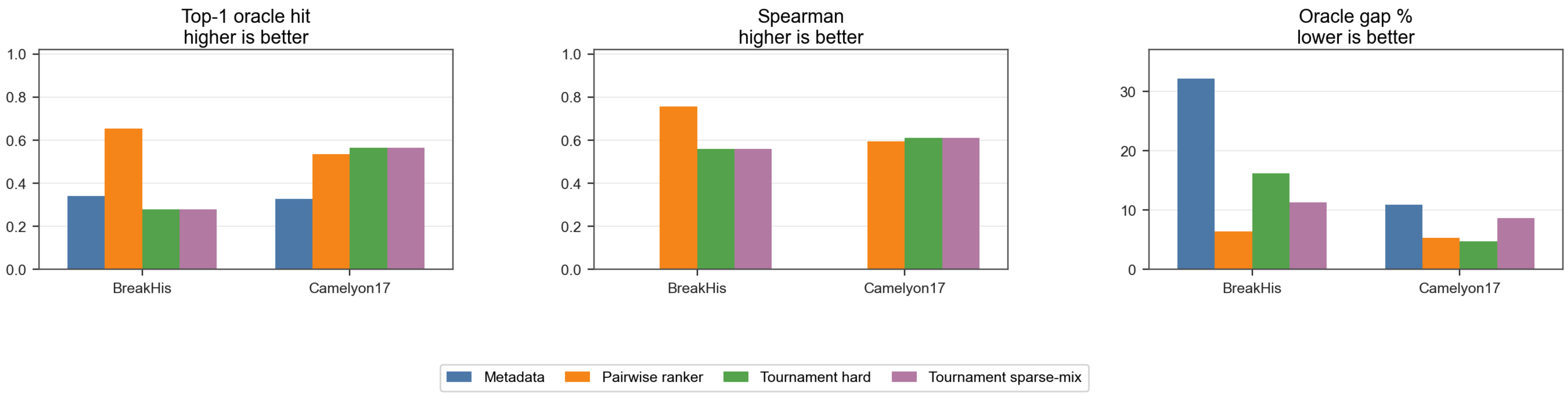
Result B

Learned utility is not yet adoption-ready



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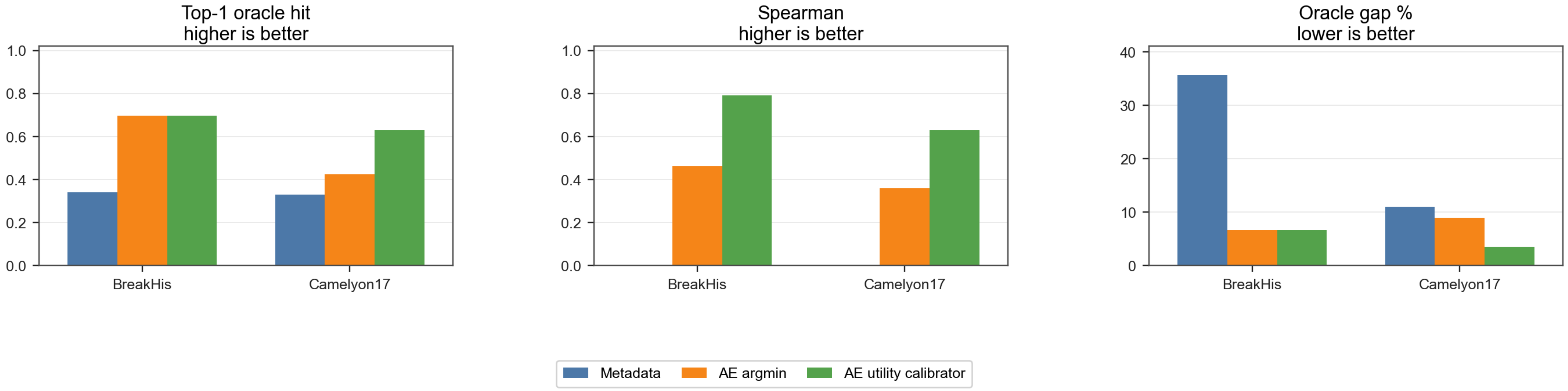
Result C

Support-free AE utility calibration



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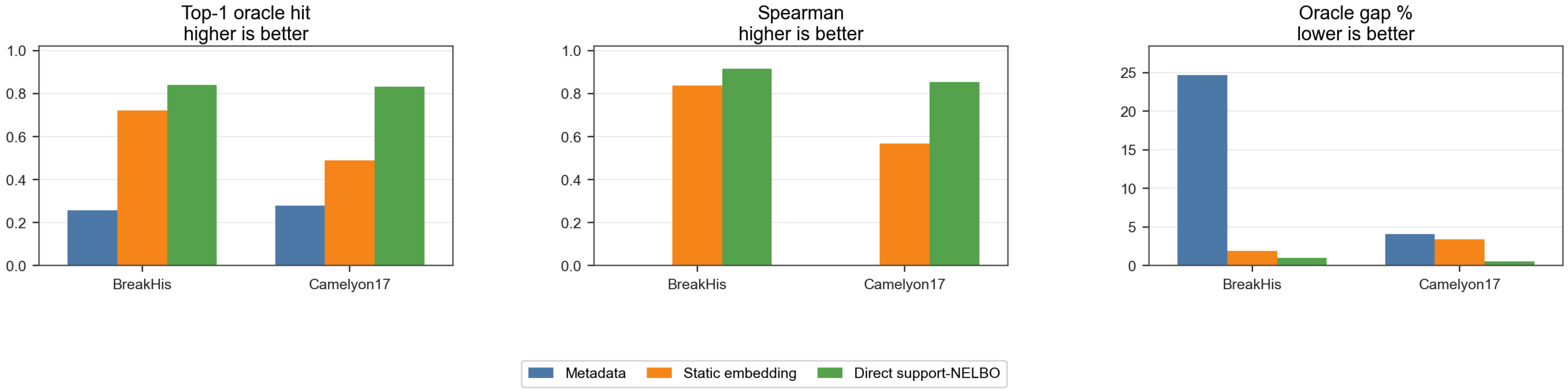
Result D

Direct support-NELBO is adoption-eligible



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Key takeaways

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Key takeaways

Hypothesis	Methods	Outcome
Domain identity is enough	Metadata routing	no
Source-only utility proxies are enough	Learned utility, pairwise tournament, AE confidence, AE calibrator	Useful but unstable
Target-local utility is needed	Direct support-NELBO	Strongest thesis-facing result

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Outlook

Thank you!